

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos 6x}{x^2} &= \lim_{x \rightarrow 0} \frac{2 \sin^2 3x}{x^2} = \lim_{x \rightarrow 0} \frac{\sin^2 3x}{\frac{x^2}{2}} = \lim_{x \rightarrow 0} \frac{\sin^2 3x}{\frac{1}{18} x^2} = 18 \cdot \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \right)^2 \rightarrow 18 \end{aligned}$$

$\left(\frac{0}{0} \right) \nearrow$

$\lim_{x \rightarrow 0} \frac{\sin x}{x} \rightarrow 1$

$\downarrow 1$

$$\lim_{x \rightarrow \infty} x(\sqrt{x^2+1} - x) = \lim_{x \rightarrow \infty} x\sqrt{x^2+1} - x^2 =$$

$$= \lim_{x \rightarrow \infty} \sqrt{x^4+x^2} - x^2 = \lim_{x \rightarrow \infty} \frac{(\sqrt{x^4+x^2} - x^2)(\sqrt{x^4+x^2} + x^2)}{\sqrt{x^4+x^2} + x^2} =$$

$$= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^4+x^2})^2 - (x^2)^2}{\sqrt{x^4+x^2} + x^2} = \lim_{x \rightarrow \infty} \frac{x^2}{\sqrt{x^4+x^2} + x^2} \left(\cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} \right) =$$

$$= \lim_{x \rightarrow \infty} \frac{1}{\frac{1}{x^2} \sqrt{x^4+x^2} + 1} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x^2}} + 1} \rightarrow \left(\frac{1}{2} \right)$$

0

$$\int_5^{12} \frac{x}{\sqrt{x+4}} dx \quad (\cong)$$

$$t = x + 4, \quad x = t - 4$$

$$dt = d(x+4) = (x+4)' dx = dx$$

$$\begin{aligned} \int \frac{t-4}{\sqrt{t}} dt &= \int \frac{t^1}{t^{\frac{1}{2}}} dt - 4 \int t^{-\frac{1}{2}} dt = \\ &= \frac{t^{\frac{1}{2}+1}}{\frac{1}{2}+1} - 4 \cdot \frac{t^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = \frac{2t^{\frac{3}{2}}}{3} - \frac{8t^{\frac{1}{2}}}{1} = \frac{2t^{\frac{3}{2}} - 8t^{\frac{1}{2}}}{3} = \\ &= \frac{2(x+4)^{\frac{3}{2}} - 8(x+4)^{\frac{1}{2}}}{3} = \frac{2(x+4)\sqrt{x+4} - 8\sqrt{x+4}}{3} = \\ &= \frac{\sqrt{x+4}(2x-8)}{3} = \frac{2(x-4)\sqrt{x+4}}{3} + C \end{aligned}$$

$$\begin{aligned} (\cong) \quad \left. \frac{2(x-4)\sqrt{x+4}}{3} \right|_5^{12} &= \frac{2(12-4)\sqrt{12+4} - 2(5-4)\sqrt{5+4}}{3} = \\ &= \frac{2 \cdot 4 \cdot 4 - 2(-3) \cdot 3}{3} = \frac{32 + 18}{3} = \frac{50}{3} \end{aligned}$$